

# Internetwork futures

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*This paper describes the architectural principles and drivers behind the Internet protocol suite and identifies the fundamental differences when compared to the historical telco approach. The major Internet protocol enhancements of multicast routing and quality of service (QoS) are then described and briefly compared to the related and developing solutions in the ITU and ATM forum. Finally, two major interworking scenarios are discussed for the Internet. The first outlines some of the fundamental architectural issues being addressed for the development of ATM within the internetworking paradigm. The second considers opportunities for the sharing of intelligence and data with the intelligent network.*

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## 1. Introduction

Traditionally, the key barriers to rapid expansion of any set of services were the cost and time-scales associated with the network of switches, bandwidth and support systems required to be developed and installed to interconnect and manage the service elements. This could not begin until the design and standardisation of the associated communication protocols, for transporting and signalling user services, had been completed. The Internet, however, has rapidly gained a global presence. The key reasons for the rapid growth of the Internet lie in its development history and resulting internetworking architecture. This enables the Internet to run over any installed network infrastructure such as leased-lines, the PSTN, X.25 data networks, frame relay, Ethernet, FDDI, ATM, cellular networks and satellites, ensuring global reach and rapid deployment. The Internet can also quickly deploy global services by distributing new application and middleware software using existing and reusable communication protocols. Its rapid expansion is therefore not simply a result of the popular data applications such as e-mail and the World Wide Web, which are merely beneficiaries of the consequences of information networking connectivity. So how did this architecture develop and what makes it so different from the telecommunications networks which preceded and presently underpin the global Internet? And how will the Internet develop in the future from the rapidly expanding, yet immature, multimedia experiment which is how many commentators appear to view the Internet today?

## 2. Protocol architecture and historical development

### 2.1 The bomb-proof network

The seven-layer OSI stack defines the functional separation between the physical means to connect elements and the user application. This discussion focuses on the lower four layers, namely the physical, link, network and transport layers. The Internet protocol (IP) suite [1] grew out of military research which had as its primary objectives the definition of a means to communicate across any type of existing link and physical media. Once communication was established, it would have to be resilient to the failure of either the links or the switching nodes as the operation of these elements could not be guaranteed in a war situation. The network protocol was literally designed to be bomb-proof. These requirements resulted in the definition of 'an internetwork protocol'<sup>1</sup> which could run over any network and is embraced by the phrase 'IP over everything'. This protocol was connectionless in nature, allowing any packet between two hosts to take a different path across the network.

The fragility of the network would necessitate using techniques to avoid fate-sharing, to ensure that the failure of a network element could not affect the end terminals and

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<sup>1</sup> A paradigm attributed to Vint Cerf of MCI.

vice-versa. This technique became known as soft-state, whereby explicit user information (state) is only temporarily stored in network elements and must be refreshed to remain in existence. Soft-state, which is still the norm in Internet Engineering Task Force (IETF) protocol design, is in contrast to hard state which is used in circuit-switched networks such as the PSTN and ATM, where an explicit set-up and disconnect signalling system is used to create and destroy network state and end-to-end connectivity.

In such a fragile environment, a transport protocol is needed to re-order packets which arrived via different paths, or routes, to re-transmit packets lost in transit, and to undertake flow control to protect the network, and other users, from congestion. An additional consequence of internetworking, connectionless routing and fate-sharing was the need to keep the network elements very simple and not rely on them to undertake flow control or retransmission on a hop-by-hop basis. This was because the traffic from a host might not traverse the same hop each time, that some networks and links perform these functions and some do not, and that network elements cannot be relied upon, through mis-operation, to provide loss-less data delivery in a controlled manner. This design decision, known as the 'end-to-end principle', asserts that only a transport protocol, running end-to-end between the two communicating hosts, can be relied upon to deliver ordered, error-free data in a controlled and fair manner. It also asserts that, in general, the network should be responsible for routing only and that the hosts should be in charge of control.

Within the Internet protocol suite, it is the transmission control protocol (TCP) which provides the reliable transport service, and the end-to-end principle dictates that its flow control must be exemplary because the network does not have its own flow control to protect itself. A simpler user datagram protocol (UDP) is used for unreliable transport for traffic such as domain name service (DNS) and real-time streaming as described elsewhere in this issue. The end-to-end principle has been extended in recent years through the concept of 'application layer framing' (ALF) [2]. This recommends that only the application is aware and able to effect control, through selection of frame size and lower layer protocol requirements, and that the selected lower layer protocols should simply provide a set of generic application services. In many respects, ALF recognises the fact that the end-to-end principle is moving up the stack, suggesting that lower layers must be simple yet controllable to enable more diverse and dynamic application requirements to be supported in the Internet. In particular, the real-time transport protocol (RTP) [3] is a new framing protocol used for transferring time-sensitive traffic (video, audio and data). In adhering to the principles of ALF, it provides fields to insert time-stamps and sequence numbers, identify the media and coding type, and select mixing and source aggregation within the internetwork. The principles

of ALF and the end-to-end principle must now be applied, where appropriate, to the multicast, reliable multicast and quality of service (QoS) protocols being designed for the Internet, which are considered in section 4.

## 2.2 *Internet growth*

The academics working on computer networking realised that they could benefit from the design decisions employed for the IP protocol. Firstly, they could build an internetwork to connect themselves using existing network technologies, avoiding the need for a direct and dedicated telecommunications connection. They could bypass any technology interworking problems and achieve multiplexing gains by using a packet gateway with network specific interfaces to route the internetwork, connectionless packets between each other. When they went home, or when they wanted to upgrade software to try something new, they could turn off the router and the routing protocols would re-route around that node as in the military requirement. The gateway became known as the router which could switch IP packets. Once connected, useful services such as Telnet, e-mail and file transfer could be deployed, fuelling the benefits of being connected to this network. Good performance across a network of networks was possible because these data applications placed no stringent requirements on the router interconnections. Rapid growth and globalisation resulted by utilising installed networks and upgrading the protocols and routers to provide sufficient addresses, routing capability and additional functionality.

The new network functionality, when deployed in all switches, produced new end-to-end capabilities. This rapid growth did, however, force good functional architecture, design and implementation, as well as rapid development (standardisation) processes. For example, deploying protocols as stand-alone processes, in the hosts and routers was very inefficient, and a different approach, known as 'integrated layer processing' is now used. Another principle of protocol design that grew out of this rapid change, and the need to maintain connectivity, is attributed to John Postel, of the Internet Assigned Numbers Authority (IANA): 'Be liberal in what you receive and conservative in what you send'. This is common sense, and can be readily applied to human communication by observing that we always try to understand what someone else says, while trying ourselves to speak as clearly as possible. It was, however, in stark contrast to stricter approaches of protocol being supported at the time. The rapid growth of the Internet has continued exponentially since those early days and has been best described and explained by the catch-phrase, attributed to Anthony Rutkowski of the Internet society, that '...connectivity has its own reward'. Connectivity is the key application of the Internet, opening up a vast range of useful applications which can be rapidly distributed and utilised

because people are connected. This should not be so strange to telcos because the historical benefits of the telephone were clearly magnified by the number of people who connected, not only leading to economies of scale for the providers and reduced entry costs for users, but also increasing the uses to which the telephone could be put.

### 2.3 The IETF

The Internet protocol suite is now described by a set of global standards produced by an informal apolitical body called the Internet Engineering Task Force [4]. This body is a volunteer organisation which has open individual membership, with the majority of the work completed via e-mail. The IETF technical working groups make decisions through consensus forming, with engineering excellence and interoperability as the major considerations. This philosophy was summed up in 1992 by Dave Clark at MIT as: 'We reject kings, presidents and voting; we believe in rough consensus and running code'. At the last IETF it was comically postulated on tee-shirts that the OSI seven-layer model should be extended to 9 layers, to embrace finance and politics. This is symptomatic of the growing pains which are so evident in the IETF and the Internet as a result of its growing importance. If the Internet protocol suite has an architecture, then it is best described by the quotations in the above paragraphs which have served the Internet well in its brief history. The challenge now is to respond to the commercial demands of the multimedia revolution by developing new network capabilities and by effectively merging the communications and standardisation models of the computer networking and telecommunications industries.

### 3. Comparison to telco networks

Telecommunications services have traditionally been intimately associated with a particular network transport technology as seen in Fig 1. This is because the associated standards process normally started with a service perspective for existing networks or both a service and technology perspective for new networks. These led to the definition of closely coupled signalling and network technology standards (such as protocol stacks) to support those services, based on a prediction of user's needs. The signalling interface to the user would offer a fixed portfolio of capabilities and only standardised services could be handled or manipulated in the network. This clearly limited innovation, because the network could only identify different services through the standardised signalling information. The alternative would have been the removal of legacy equipment which for major global infrastructure like the PSTN is clearly not viable. Upgrading the network would take time and would need to avoid incompatibilities and maintain backwards compatibility and any new network would require network- and service-specific gateways to be

deployed to maintain connectivity to the installed base. These application gateways would be expensive to develop because they would generally need to interconnect the physical media as well as both user-service and signalling layers, and one would be required for each combination of service technologies. The justification for this approach was that customers would want the new services which would enable the supporting network to be upgraded or superseded. Historical examples can be readily seen in the evolutionary relationships and interworking between telephony, facsimile, conferencing and data services on the ISDN, intelligent network, PSTN and X.25 networks.

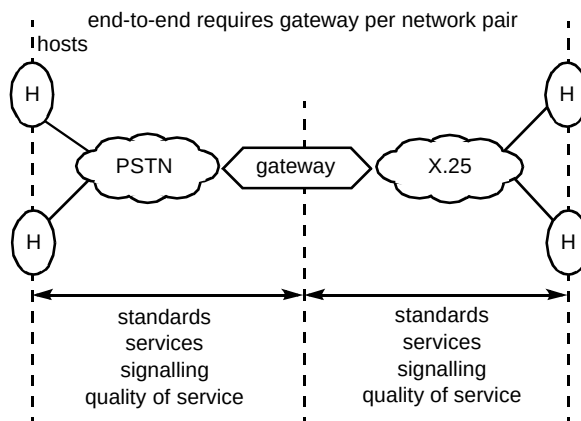


Fig 1 Traditional network architecture.

This approach, based on the development of standards which were tightly coupled to networks, technologies and services, was well suited to a monopolistic environment managed by the dominant public operators. This produced some very successful network and CPE-specific services with optimised and predictable end-to-end performance and associated pricing regimes. The approach has two major benefits from operator and customer perspectives. The network and services are designed together to guarantee end-to-end performance and the business case for network investment is supported by a business case for the associated service penetration and revenue forecasts. The disadvantages are that the service-specific elements of the network, and by implication the network itself, can only be justified with solid service projections. In an age of deregulated services and high service dynamics, this approach can be problematic and the trend is therefore swinging towards network-independent services and generic-network capabilities so that the service elements can be reused over existing and future networks and the networks can evolve gracefully with the changing demands of the user applications. The interface between network-independent services and service-independent networks is the internetworking protocol which, as a consequence, minimises investment risk both in network technology and service-specific infrastructure. IP is one such internetworking protocol which has survived and evolved as the

evolutionary winner, over similar protocols, due to a number of factors. The Internet protocol suite provides generic communication capabilities allowing the Internet to utilise any installed or planned network technology as shown in Fig 2.

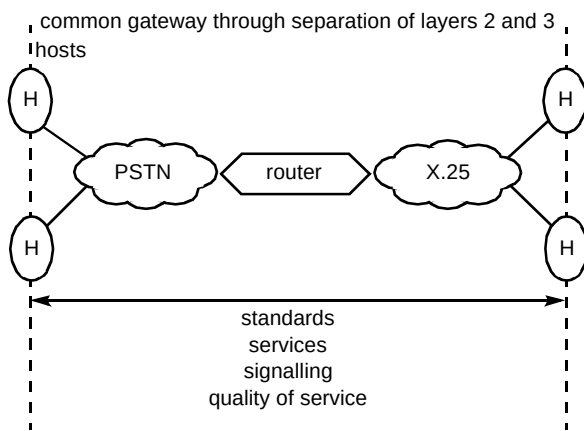


Fig 2 Internetworking architecture.

Only a single network-layer gateway is now required because the network signalling and physical characteristics are isolated from each other, and replaced by gateway-to-gateway, host-to-gateway and host-to-host signalling, carried by (encapsulated within) the internetwork protocol. In particular, application control and capability negotiation is now compliant with the end-to-end principle, and the underlying networks simply provide bandwidth and connectivity.

In contrast to traditional telecommunications platforms, the Internet is oriented towards generic internetwork capability rather than specific user services, the former being the norm within the computing industry. The associated service environment is generally unregulated, non-standard and both highly dynamic and very competitive. A good example is the PC where the user buys capabilities such as CPU, memory, hard drives, monitor, accessories and operating system to open up access to a whole range of proprietary applications and services, instead of buying application specific terminals such as a telephone or a video-conferencing unit. The role of Internet standards is to ensure that connectivity across an environment of heterogeneous network technologies and topologies is supported between heterogeneous terminals using a suite of internetworking protocols. Some services are also standardised because they are essential for this internetworking capability (e.g. DNS), the impact on the network of proprietary services is damaging (e.g. TCP), or the service itself becomes a fundamental component of the capability (e.g. SMTP). Network capabilities are only deployed if the impact on the heterogeneous internetwork is meaningful, scaleable, manageable and beneficial. Useful capabilities are identified as part of an ongoing process to

measure how the Internet is performing, what services are being used, what services may be around the corner and what potential capability might stimulate a new market. The Internet protocol suite has identifiable protocol and port numbers which can be detected both end-to-end and in the network to aid this process. This can be viewed as a very flexible generic signalling mechanism. In particular, the port numbers can be allocated to new applications which can be used to access application-specific processes. Certain port numbers are allocated as part of standardised services (WWW, DNS, e-mail), but a huge number of ports exist for customised applications and access to value-added processing and functionality in the network.

The Internet protocol suite is designed within a functionally oriented architecture. It has some similarity with the OSI seven-layer model but is more flexible in its interpretation of layering. The aim is to achieve a good functional decomposition of responsibilities and capabilities resulting in a set of protocols that provide generic functionality which can be reused by other protocols within the architecture. Good design minimises the impact of protocol change on other network protocols and the network elements so that rapid evolution, through protocol standardisation, can result in additional network capability. In particular, protocols can undertake their own signalling and/or rely on other protocols rather than depend on closely-coupled network and application signalling stacks. Recently, the IETF has decreed that each protocol standardisation process also addresses security and management implications and capabilities so that the associated issues are segmented and solved on a 'per protocol' basis, and can quickly lead to useful, deployable capability. This layered functional approach maximises flexibility at both the network and service layers and ensures that all useful functionality is exposed to protocol and application authors.

The standardisation of the Internet protocol suite is therefore very open and many implementations exist in the market as a result. The focus on interoperability in protocol design, standardisation and implementation ensures a healthy market for interworking hardware and software products, and vendors compete through the quality of their implementations and business processes. This open architecture actively encourages hardware and software developers to add value to the standardised protocol architecture, which has led to much innovation in its relatively short life. The end result is a platform which can evolve quickly in response to market requirements and which can migrate suitable generic functionality from the software layers into the network elements (multicast, mobility, intelligence), creating an environment in which anyone can write the key application (Netscape, Mosaic, search engines). Proprietary services exhibiting high rates of change do have a side effect, namely that great skill must be applied to identify the stable and profitable services

which benefit both the user and operator and which can therefore have service and billing processes defined, for them to manage and fund those services. One solution is to concentrate instead on building such systems around the generic protocols and network capabilities, relying on charging for connectivity and capability employed rather than applications themselves. This leaves application writers to profit in a similar way to the evolution of the PC market, but it raises uncertainties regarding end-to-end application performance and allocation of responsibilities regarding customer support.

#### 4. Multimedia networking

##### 4.1 Multicast

One of the new networking capabilities which has grown out of developments in data networking and the Internet is network multicasting. Multicasting is a generic term which describes how information is distributed among a participating group — compare other distribution types:

- unicast — one-to-one information transfer, which is used for the most traditional form of telecommunications, with standard telephony comprising two associated unicast paths,
- broadcasting — one-to-all distribution, as exemplified by radio or television broadcast, the distribution being to 'all', irrespective of whether receivers are listening,
- multicasting —  $n$ -to- $m$  distribution, where the receivers, ( $m$ ), are an identifiable group of potential 'listeners', an example of multicast distribution being an audio conference where everyone in the conference group hears everyone who is speaking (both one-to- $m$  and one-to-all (broadcasting) are special cases of multicasting).

Multicasting capability may be provided in a number of ways. Traditional link layers such as broadcast LANs and satellite links provide a natural means to distribute a single piece of information efficiently to multiple receivers. This can be converted to a multicast service by enabling the receiving link layer interface to only forward incoming traffic from requested multicast groups.

Multicasting can also be achieved at the application, session and transport layers. Examples include the forwarding of e-mail to multiple recipients, where e-mail applications receive, replicate and forward the message. In the general case, senders transmit (unicast) their message to a multicast application, which reconstructs and interprets the message sent by the original application. The application performs any replication and forwards the message to the destinations. The application forwards messages using the point-to-point service of the underlying

unicast network although an underlying multicast network service may be used.

Higher layer multicast has the advantages that it can be provided with no modification to the underlying networks, and the multicast application layer may be used to provide more complex control and processing functions on the data. The drawbacks of this approach, however, are that the multicasting and control functions are application-specific in that the multicast applications must be able to understand the format and semantics of the original messages. In addition, there is little saving in network bandwidth (with duplicate copies of data being sent over the same links), unless a significant network of geographically distributed multicast applications is deployed. This can be seen in Fig 3.

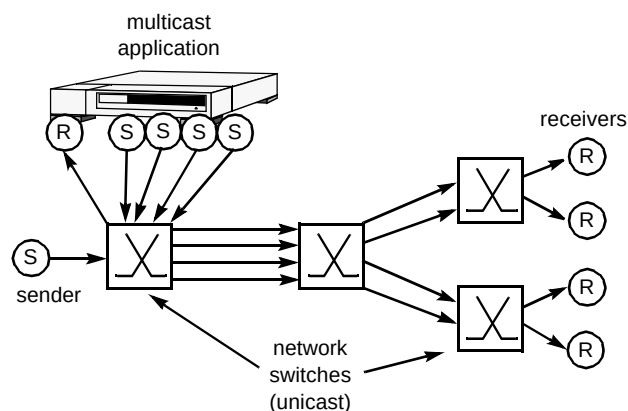


Fig 3 Application layer multicast.

Traditional telco switched networks have used this approach to provide multipoint services. This can be attributed to the connection-oriented paradigm which provides circuits between multipoint terminals at the edge of the network supporting higher layer (e.g. transport) functionality. The provision of audio-conferencing over the ISDN is a good example of this, where an audio bridge is provided by the operator to bring together the multiple unicast connections, mix the audio content and distribute to all participants. A dynamic session-layer identifier is used to associate the E.164 addresses of the participants within the multicast group, and either the bridge or the participants must dial up the appropriate connections to form the multicast tree. This connection-oriented approach can be improved, as in the ATM standards effort, using network-based multicast servers to undertake replication and group management, and here the multicast group address and call control is handled by the initiator of the group which is typically the sender. Tree building is then an automatic function

Developments in the ITU have looked at more generic multicasting within the T.120 umbrella standard for data-sharing applications, whiteboards and associated conference

control. T.120 provides the ability to dynamically build and maintain application layer multicast trees and is also able to run over a number of different networks. Extensions are being developed to make use of underlying network and link multicast where available. This does not mean, however, that higher layer multicast will not have a significant role to play in the future but once again highlights the different architectures which need to be embraced, including the architecture being developed by the IETF.

### IP multicast framework

The philosophy behind Internet multicasting is to provide a generic multicasting capability irrespective of the type of data or the application which created it. The same multicast group can therefore be used for voice, video and data.

A dynamically allocated network-layer multicast address is used to identify the multicast group. The network elements (switches or routers) recognise the multicast address and replicate and forward the datagram along the distribution tree. This tree is constructed by dynamic, connectionless multicast routing protocols, many of which rely on the underlying unicast routing protocols. The multicast protocols aim to ensure that only a single copy of each datagram is forwarded between any two network elements, including user terminals, as shown in Fig 4, irrespective of the number and position of downstream receivers. This is particularly beneficial for the global Internet, providing efficient use of international and national leased lines and decoupling the servers from the clients. The latter ensures that only a single stream has to be delivered by each server. Hence, the usefulness of the multicasting capability to the network operator is in far more efficient use of network resources, to the service provider is in reduced service provision costs, and to the client is in reduced access costs and greater control. The reader is referred to Semeria et al [5] for an excellent overview of IP multicast.

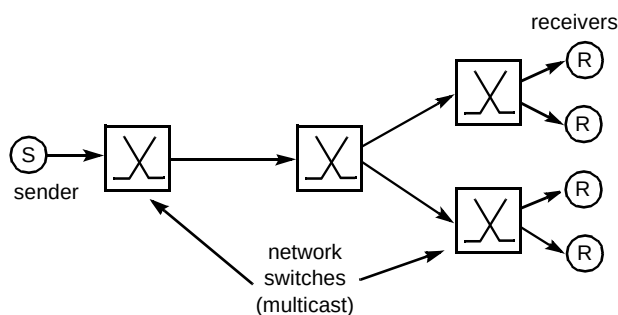


Fig 4

### IP multicast addressing

The only difference between a unicast IP packet and a multicast IP packet is that the latter carries a 'group address' in the destination address field. A range of the IPv4 address space (Class D) is used to identify multicast groups. These addresses bear no physical or topological significance but are merely logical identifiers representing a dynamically changing group of subscribed hosts. One reason for the adoption of logical rather than physical addresses is greater flexibility. This is similar to the flexibility that 0800 or 0345 numbers provide in the PSTN. In IP multicast, this allows for the capabilities of both dynamic group membership, and many-to-many connectivity. Any host is free to subscribe or unsubscribe from any group at any time, with a higher layer protocol potentially allocating and announcing multicast addresses and associated session, content and membership. It is, however, current practice to restrict the scope of multicast data. This may be done either by using datagram hop-count limits or by setting administrative boundaries for certain multicast address ranges.

### Multicast group management and routing

The IP multicast framework requires that routers, but not hosts, run routing protocols. This means that a separate protocol is needed to communicate group membership between hosts and their last-hop (local) multicast router. This is the function of the Internet group management protocol (IGMP). Once the host has notified the local multicast router of its group membership, the router can participate in multicast routing to ensure that it is added to the distribution tree. As the number of subscribing members increases, the possible permutations in the formation of a distribution tree become extremely numerous. Here, a number of design choices exist, with inherent compromises, resulting in the development of several multicast routing protocols within the IETF. These are distance vector multicast routing protocol (DVMRP), multicast extensions to open shortest path first (MOSPF) (proposed standard), core-based trees (CBT), protocol-independent multicast with its sparse mode (PIM-SM) and dense mode (PIM-DM) versions [5–7]. It should be noted that only one of these has reached the status of 'Proposed Standard', the others being experimental.

There are two main design choices which distinguish multicast protocols — the tree formation mechanism, and the distribution tree type. As a result, the various protocols can differ significantly in terms of bandwidth efficiency, path delay and router overhead.

- Tree formation — broadcast-and-prune or explicit join.

Two main mechanisms exist to form the distribution tree. In broadcast-and-prune, each sender initially broadcasts datagrams throughout the multicast domain.

On receipt of the datagrams, all last-hop routers with no attached hosts subscribing to the group then send a 'prune' message upstream, thereby removing the unwanted branch of the tree. This approach is well-suited to dense-mode multicasting, i.e. where the density of subnets subscribing to the multicast group is high. Clearly, with a sparse population (low subscriber density), this incurs considerable wasted bandwidth.

Explicit join approaches were therefore developed in which data would not be forwarded to last-hop routers unless they explicitly joined themselves on to the distribution tree. So, unlike the data-driven approach of the broadcast-and-prune protocols, branches would have to be formed in advance of data being forwarded. The mechanism for last-hop routers to explicitly join the tree could be by means of a 'join' message towards the core router in a shared protocol (e.g. PIM-SM, CBT), or a broadcast membership advertisement (MOSPF).

- Distribution tree type — source-based or shared tree.

With source-based tree protocols, the forwarding path for a group is formed from an overlay of distribution trees. Each source specific tree is generally constructed using reverse path multicasting which calculates the shortest path back to the source. Branches of the tree are pruned off if not terminating on a subscribing subnet and datagrams are only accepted and forwarded if received along the remaining branches. The main strengths of the source-based tree approach are the simplicity of tree formation, and the guarantee of shortest paths. In Fig 5(a), the last-hop multicast routers R1 — R8 have directly connected hosts (not shown). However, for routers R1 and R3, none of their directly connected hosts are subscribed to the multicast group. The group has senders attached to R2 and R8 and therefore has forwarding trees rooted at R2 and R8.

With shared tree protocols, a single distribution tree is used for all sources. This tree is formed around a core router which acts as the root of the tree (or, the centre of a star, from a different viewpoint); senders transmit towards the core router and the datagrams are replicated as they travel away from the core router to the receivers. By its nature, use of a shared tree cannot guarantee shortest paths between all senders and all receivers, but has the advantage of requiring less overhead than a source-based tree approach particularly when there are many sources and the tree is bi-directional. A shared tree is shown in Fig 5(b) which now has a single forwarding tree for all sources.

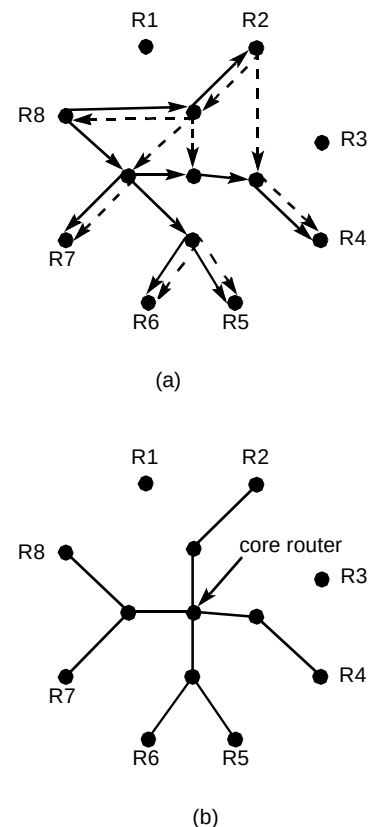


Fig 5 Multicast forwarding trees.

### Reliability

IP multicast provides connectionless forwarding, without guaranteed delivery or sequencing. This is perfectly adequate for audio and video distribution, and even data distribution where a small amount of data loss is acceptable. Data loss for audio and video applications can be avoided using appropriate dimensioning or by adding QoS protocols as discussed in section 4.2. To ensure reliable delivery of data, a reliable multicast protocol must be added at the transport layer, above multicast routing. Such a protocol therefore bears the same relation to IP multicast as does TCP to IP unicast. Historically, and despite its limitations, TCP has provided a very widely applicable reliable unicast transport service, and the protocol is relatively simple and robust. The same is not true of reliable multicast for two reasons:

- the application requirements for reliable multicast are far more diverse than for reliable unicast,
- satisfying the basic requirements of reliable transport and congestion control is made extremely complex when there are multiple receivers.

Basically, the communication model as a 'conversation' breaks down, but it is clearly essential that reliable multicast protocols include an effective form of congestion control, to safeguard the Internet.

In terms of retransmission of lost data, a design decision has to be taken as to whether repair of the lost data is undertaken by the original sender, a network element or a local client who successfully received that data. Application requirements for a reliable multicast protocol may also include, for example:

- levels of reliability — for different data types or for different receivers,
- data naming — transport data unit identifiers in a multi-sender environment,
- ordering of data — unordered, source ordered, or totally ordered (repair before or after new data?),
- timeliness — for initial and repair transmission,
- resilience and atomicity — how the session is to respond if sites fail or partition.

As a result of the diverse requirements, it has been acknowledged that a 'one-fits-all' protocol may not be possible and consequently a large number of proprietary reliable multicast protocols have been designed, including multicast file transfer protocol (MFTP), scalable reliable multicast (SRM), reliable multicast protocol (RMP) and multicast transport protocol (MTP) [8, 9]. Having a large number of transport protocols to choose from is clearly not beneficial to users, operators or vendors. This is particularly true when one considers the existence of the ITU T.120 series of protocols which provides this capability using a different technical and commercial paradigm. Once again there is an opportunity for a convergence of ideas and architecture.

Application-layer framing would suggest a solution based on a lightweight framing protocol which can support the major generic requirements, supplemented with diverse higher layer functionality. At a recent meeting of the IETF, it was decided to attempt to co-ordinate the diverse work on reliable multicast for the Internet, while recognising that there were complex issues and important architectural questions to be resolved. A particular question concerns the applicability of the end-to-end principle because this would imply that the clients handle repair and not the network, but as in pure multicast, significant efficiencies result if the network is involved in repair.

#### IP multicast issues

In 1992, an experimental multicast network known as the Mbone (multicast backbone) was formed. This is an overlay to the Internet — islands of multicast-capable routers connected by unicast Internet links. This has now grown to include around 2800 sub-networks (typically computer local area networks) in over 25 countries and has the capability for large-scale multimedia conferencing. It is

used, for example, to link large audiences with national and international technical conferences using audio, video and shared data whiteboards. The Mbone has seen rapid growth over the last few years due to the number of connecting sites, the number and bandwidth requirements of marketable IP multicast applications and the resulting usefulness, and hence occupancy, of the multicast links and routers. There has also been considerable experience gained in the use of IP multicast routing.

The next step in the IP multicast story will be the widespread deployment in IP networks as a native capability. Recently, a major IP network provider in the US was involved in a demonstration of wide-scale IP native multicasting. A further significant event was the formation of the IP Multicast Initiative, a consortium of many leading router vendors, Internet service providers (ISPs) and application and middleware developers, with the goal of expediting the deployment and use of IP multicast. There are, however, major outstanding issues, including:

- the lack of hierarchy in multicast routing,
- the lack of hierarchy in multicast addressing,
- the complexity and overhead of multicast protocols,
- the lack of control over tree and join characteristics,
- the lack of policy controls for ISPs.

The first three of the above issues, given the huge existing and predicted demand, pose scaling limitations to the network elements, which is being addressed by the IETF and vendors. The consequences of poor scalability, and the need for protocols to be adaptable to serve the diverse range of multicast customer and operator requirements, leads to the last two issues in the above list.

#### Multicast summary

Multicasting is a networking paradigm which is changing the way in which multi-party services are designed and deployed. The key benefits of multicasting are reduced service-provider costs and core bandwidth efficiency. The IETF, ITU and ATM Forum are now actively engaged in developing next generation multicast protocols but with major architectural differences. It may be an appropriate time for network operators to reassess the different directions and models to achieve convergence within an internetwork architecture.

The Internet model offers a generic multicast capability which can be used to deploy a wide range of services. It enables distribution of multimedia content to multiple endpoints and is a key component of future conferencing and group-working applications. The aim is to ensure that the multicast Internet will be able to support very large

numbers of multicast groups, sources and receivers within a viable commercial model. Presently, it is best suited to distribution (Internet radio, TV) and informal conferencing services due to the loose coupling at the network layer. The need for increased manual or automatic management intervention to provide sufficient control, potentially at higher layers of the protocol stack, has been recognised to support tightly-coupled conferences and avoid damaging network stability.

As a result of the success of the Mbone, multimedia clients are already available for most computing platforms and they provide access to efficient multicast distribution mechanisms. These clients and the internetwork will, however, require a reservation mechanism if end-to-end QoS guarantees are required. These are also being standardised by the IETF and are discussed below.

## 4.2 Integrated services and RSVP

### Best-effort Internet

Currently, the Internet does not provide any guarantees of quality of service in terms of throughput or delay experienced by data packets between source and destination. Moreover, packet delay and loss are influenced by the instantaneous volume of traffic through any particular router. For example, a lightly loaded router might exhibit a packet delay of less than 5 ms, whereas at a busy point in the network the delay could exceed 100 ms. Given that traffic often passes through many routers there is a good chance that delay variations of orders of magnitude could be experienced. This was acceptable for the traditional data applications and a mixture of suitable access controls and dimensioning can readily provide sufficient performance. This may not, however, be adequate for delay sensitive services such as voice or video, as it leads to significant variations in quality within the session.

Within a single telecommunications network technology, the provision of QoS is a relatively simple task in that connections of the required capability can be set up and maintained for the duration of the session. Dimensioning is required to ensure that signalling and switching network elements can provide a high probability for dial-tone and connectivity. Note that end-station busy tone is not considered to be a failure. In the Internet, the only real difference is in the dimensioning and congestion failure mode. The public Internet today is a cheap and cheerful service necessitating tight dimensioning constraints. The access controls are in the form of the number of modems, the size of the access pipes and the limited number of server ports. In addition, the failure mode of the Internet is very different in that a busy modem and server is considered a failure and under-dimensioning leads to lower throughput.

engaged tone and existing traffic would be preserved, but there would be equivalent levels, if not worse, of discontent from those refused access. In many application circumstances, the lack of statistical multiplexing in the network and at the end station is a key limitation which has been addressed in recent standardisation efforts for ATM. However, the real issue here is whether the Internet will generate sufficient revenue to raise the dimensioning levels to that enjoyed within the PSTN, and whether the user can negotiate and receive an agreed level of service.

### Internet requirements

Within the Internet community this is well recognised and has led to significant developments to support end-to-end QoS negotiation. This requires the addition of an internetwork protocol which can carry QoS requirements through the routers connecting two hosts. It also requires a means for those routers to use those parameters to prioritise traffic and also either allocate fractions of the bulk bandwidth between themselves, or interact with the underlying switching technology to set up appropriate connections. On shared media such as Ethernet, FDDI, cable networks and radio, it is also necessary to upgrade the medium access control (MAC) layer to enable access control and prioritisation to be implemented across the connected interfaces.

### Integrated services architecture

In the IETF, the integrated services (Int-Serv) [10] architecture defines a new service model for the Internet which accommodates both best-effort and real-time, delay-managed services. This includes the requirements for scheduling algorithms within the network elements, with weighted fair queuing and class-based queuing being the best known. The resource reservation set-up protocol (RSVP) [11] provides a means to transport integrated service QoS requirements between QoS-aware network elements and applications, as part of a unicast or multicast delivery system. RSVP is therefore a signalling mechanism for applications to request and receive preferential services, compliant with the Int-Serv architecture, through an internetwork. RSVP protocol mechanisms provide a general facility for creating and maintaining distributed reservation state across a mesh of multicast or unicast delivery paths through an internetwork. RSVP transfers reservation parameters, such as bandwidth and delay requirements, as opaque data (except for certain well-defined operations on the data), which it simply passes to a network element for interpretation. Although the RSVP protocol mechanisms are largely independent of the encoding of these parameters, the encodings must be defined in the reservation model that is presented to an application as defined by integrated services. An example host implementation is shown in Fig 6, and described below.

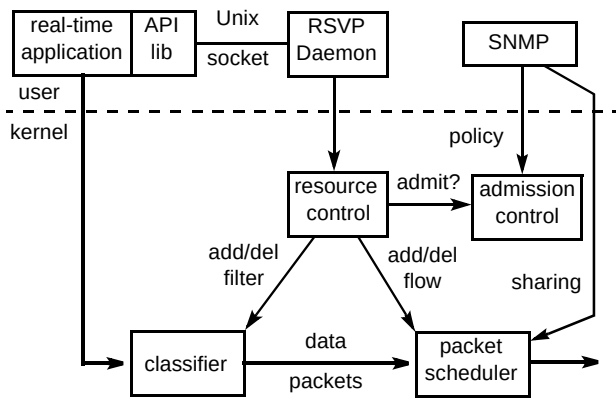


Fig 6 RSVP host architecture.

A sending application will first announce to receivers what data it will be providing and in what form. The receiving application(s) will then formulate a reservation request to ensure timely delivery of this data. This QoS request will be passed by an API to the local RSVP implementation, shown as a user daemon in Fig 6. The RSVP protocol is then used to pass the request to all the nodes along the reverse data path(s) to the data source(s). Suitable priority and management functions will then be applied to the sender's data as it traverses the route back to the receiver. In particular, the receiving host must undertake admission control to ensure its local configuration and network interface can support the request. Policy decisions must be made to restrict reservation functionality to identified receivers through the use of the simple network management protocol (SNMP).

Each node (hosts and routers) that is capable of resource reservation passes incoming data packets to a packet classifier and then queues them as necessary in a packet scheduler. The packet classifier determines the route and the QoS class for each packet. The scheduler allocates resources for transmission on the particular link-layer medium used by each interface. If the link-layer medium is QoS-active by virtue of its own QoS management capability, then the packet scheduler is responsible for negotiation with the link layer to obtain the QoS requested by RSVP. There are many possible ways this might be accomplished, and the details will be medium-dependent. The scheduler itself allocates packet transmission capacity on a QoS-passive medium such as a leased line. The scheduler may also allocate other system resources such as CPU time or buffers.

RSVP also defines the processes for combining and controlling heterogeneous QoS requests and resources for multicast applications. The protocol is designed to support small numbers of multicast groups per router, with each multicast group potentially supporting large numbers of users. RSVP has the following main attributes:

- support for multicast or unicast delivery,
- adapts to changing group membership and routing,
- reservations initiated and controlled by the receiver and not the sender,
- several reservation styles tailored to fit specific classes of applications,
- managed performance on a hop-by-hop basis within end-to-end constraints.

Note that RSVP has not been designed for large numbers of point-to-point connections, such as would be required for Internet telephony, a subject which has been generating much interest [12].

#### 4.3 RSVP summary and issues

RSVP and the associated Integrated Service scheduling algorithms give users preferential access to Internet resources. Therefore, some form of back-pressure must be applied to ensure that reservations are only granted to a subset of the user base. This can be achieved by charging for reservations or using security authentication for reservation requests. Both authentication and charging models and mechanisms are being developed which could lead to the upgrade of the Internet within the next couple of years, but could be hampered by the commercial agreements which must be in place to support transfer charging for cross-provider reservations.

With current Internet router technology, some performance limitations are evident for RSVP processing in hosts and routers. These limit the rate at which RSVP connections can be set up and taken down, and the load on existing router processors and memory for running the scheduling algorithms to manage QoS and performance. These routers were, however, designed for traditional data, not multimedia traffic and this challenge is being taken up by vendors who are now able to deliver gigabit performance. RSVP was designed for receiver-based reservation and to allow the routers to merge reservation requests from different receivers based on the underlying multicast topology. Reservation requests and confirmations are therefore generated at the edge and merge points and not end-to-end. This ensures that heterogeneous receivers can be supported and that the signalling messages per link will scale to large numbers of users. RSVP uses soft-state (refresh timers) to ensure that reservation state is not left unused in the network. These efficiency design decisions and the integration with multicast make both the apportionment of the cost of reservation and the QoS delivered to the user very complex issues when compared to the point-to-point connection model.

Finally, for end-to-end delay control, it is essential that the underlying network technology, between routers, can provide dedicated inter-router bandwidth which can be apportioned by the router among a number of QoS flows. This can be provided using leased lines (PDH, SDH) or a shared medium which supports QoS management such as the new IEEE link-layer standards. Legacy networks such as SMDS and Ethernet cannot actively support RSVP reservation, although appropriate traffic dimensioning can ensure resource is available by default. In addition, RSVP messages can still be carried transparently across these clouds and interpreted by connected QoS-aware networks. With a migration to ATM for public data service, a whole new opportunity for mapping IP RSVP messages into ATM signalling becomes available, providing the possibility of high-quality national and international broadband internetworking. Some of the associated issues are described in the following sections, which consider interworking issues for the intelligent network and ATM platforms.

## 5. Interworking IP and ATM

The ITU and the ATM Forum are standardising ATM as a tightly coupled network and service technology resulting in network specific capabilities for QoS (CBR, VBR), flow control (ABR), mobility (LANE, MPOA) and ATM multicast. In addition, ATM is connection-oriented and these capabilities are therefore associated with a tightly coupled signalling system. None of these characteristics are necessarily a problem, particularly if ATM becomes ubiquitous. This is certainly a possibility as ATM has a fixed frame size (ATM cell) to enable cheap and fast switching hardware and a variety of bearer classes providing customer choice in terms of tariff, transfer characteristic and end-to-end performance. The use of ATM bearers to economically interconnect hosts and routers should provide a very flexible access and core capability for customer and ISPs but many issues have yet to be resolved.

### 5.1 Functional duplication

It has been shown how IP is an internetworking protocol which aims to push the underlying networks down the OSI stack so that they merely provide connectivity and bandwidth. If ATM is used simply to provide point-to-point bandwidth then all is well and the internetwork architecture is not compromised. It has also been shown how the IETF is now adding multicast and QoS capabilities to the wide range of internetwork capabilities, which also includes mobile IP and other roaming services. These provide internetwork terminal and user mobility. When compared with the rich set of ATM functionality which is now available, it begs the question as to whether both IP and ATM can implement and interwork all this functionality. If not, then which functional model is correct and which layer

should handle each function? Is it the end-to-end, window-based flow control of TCP or the network-specific rate-based flow control of ABR, and can they coexist and interwork? Is it soft-state or hard-state? ATM-based statistical multiplexing and cell multiplexing is in competition with IP packet multiplexing over datagram-oriented networks and links such as frame relay and SDH. Who is now responsible for routing and customer performance with the coexistence of sender-oriented ATM multicast/QoS signalling with the receiver-based model of the IETF? And what about network-specific mobility through LANE and MPOA as compared to the internetwork mobility of mobile IP and roaming services?

### 5.2 Connectionless or connection oriented?

The ubiquitous multiservice, cross-platform nature of the Internet is very much in its favour. In addition, a connectionless environment may well be better suited for future networks because it is the lowest common denominator for multimedia, multiservice applications. Also, all the trends point to data being dominant over real-time traffic, and this data traffic may be characterised by higher volumes of transactional sessions with disparate durations. On the other hand, the customer's traditional emphasis on performance, service management and end-to-end QoS guarantees may well be better served by connection-oriented networks. For example, IP over a PSTN connection can offer guaranteed bandwidth and performance end-to-end. Similarly, ISDN, frame relay and ATM gateways can be used to break-out from the IP layer and fully utilise the underlying network capabilities which implies communication between the IP layer and the underlying network signalling. However, signalling and connection set-up should only be invoked for longer, identifiable IP sessions, with stable QoS and performance requirements. This is to avoid overloading the connection-oriented network signalling and affecting traditional services. An alternative would be to support multimedia and QoS services by running connection-oriented IP signalling such as RSVP. In effect, RSVP within SDH-connected routers can be considered to be in competition with the deployment plans for new network technologies such as ATM.

### 5.3 Convergence

ATM signalling offers a rich capability but has been defined over many years to support traditional service models. It is not necessarily the best solution for internetwork vendors who wish to make use of the ATM fast-switching technology. Products are therefore appearing from companies such as Ipsilon, Toshiba and Cisco, both for the LAN and WAN, which to a greater or lesser extent merge the IP protocol architecture and ATM technology

within a single element, attempting to avoid the interworking complexities between the IP and ATM network functionality due to signalling and functional duplication. This might leave ATM networks to provide bulk inter-router bandwidth and flexible, cost-effective point-to-point connections for Internet access, and other services. The key issue for the industry, therefore, is to maintain the integrity of the various standards and installed platforms while exploring, both technically and commercially, the various scenarios for interworking, convergence and value-added functionality. These issues are again being addressed by a number of working groups in the IETF and the ATM Forum, and keeping a lot of design and evolution planners very busy.

## 6. Intelligent networks and the Internet

The intelligent network (IN) concept has been a major driver in the evolution of telecommunications networks during the last decade with most major telcos deploying IN-compliant platforms. Within BT, IN activities include the deployment of two service control points (SCPs) and the upgrade of a number of exchanges to support IN operation. The reader is referred elsewhere [13, 14] for a comprehensive introduction to the IN model.

The roots of the IN model, however, predate the explosive growth in personal computing power and the Internet occurring during the last few years. Furthermore, other papers in this themed issue have described a range of applications and capabilities that may be delivered across an Internet and that may lead to IN capabilities appearing to be increasingly limited by comparison. Given the importance of IN to the business and the associated investment levels, it is essential that the impact of these technological revolutions on IN evolution is understood. For the purposes of this analysis it is suggested that this impact will fall into two areas, namely:

- enhancing IN service and management capability,
- influencing medium- and long-term IN evolution.

A number of examples within each category are outlined below. While such brief consideration is in danger of raising more questions than answers, it remains useful in providing a flavour of a number of important opportunities and issues.

### 6.1 Enhancing IN service and management capability

The following sub-sections consider the use of the Internet and related technologies to extend IN capability. The first describes an architecture intended to facilitate access to IN data for management purposes. The second considers how new service opportunities may be enabled through interworking between IN intelligence and intelligence located on the periphery of the network.

#### Distributed data access

Figure 7 illustrates an architecture intended to support flexible and distributed access to IN service data across an Internet.

Key elements and associated benefits for this architecture include the following.

- WWW Browser clients with functionality implemented in Java. The Java code defining client behaviour is stored on the data access server as one or more Java 'applets' and downloaded for execution on the client when required. Benefits of this approach include:
  - deployment/upgrade transparency — assuming the existence of a suitable WWW browser on the data access client (DAC), there is no requirement for

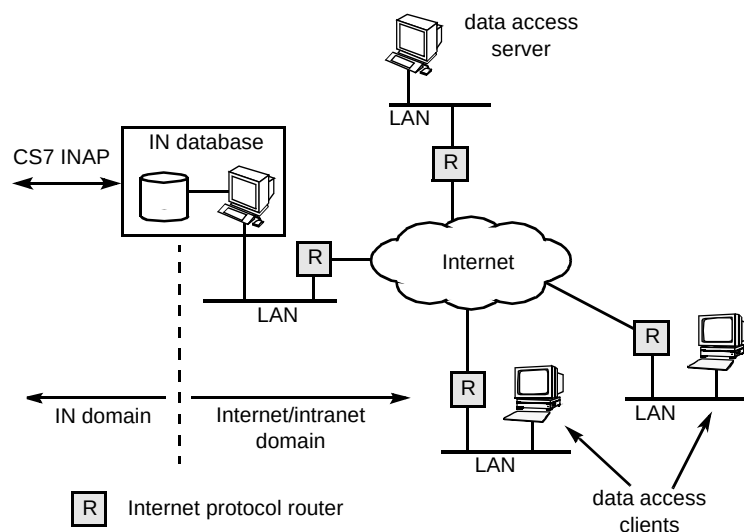


Fig 7 Distributed data access.

manual installation of service management software on the client itself, and, similarly, clients may be upgraded for repair or enhancement in the same 'hands-off' manner (new or amended applets are simply deployed on to the server, becoming immediately available to clients),

— platform transparency — the same DAC code will operate across all major operating systems, a virtual machine within the WWW browser providing the platform-independent API required to achieve this portability.

- Three-layer database architecture built on open database connectivity components. The separation of database, data access engine and user interface provides a more flexible and more easily extensible architecture. Opening up access to IN data in this manner brings benefits for both user and service provider:
  - easier access to data for operational support purposes enabling, for example, the rapid deployment of data tools to automate data build and data validation for IN services,
  - customer data access via the Internet for managing profiles, viewing billing data, etc.
- Use of object-oriented design and implementation techniques is compatible with a longer term evolution towards more distributed co-operative architectures defined in initiatives such as TINA-C. Furthermore, it is suggested that this architecture coupled with the less demanding non-real-time performance requirements of the management layer (cf. service and call control layers) provides a low-risk route to gaining operational experience of distributed object technologies which remain to be proven in terms of performance and scalability.

#### Interworking intelligence

There are a number of opportunities and issues related to interworking between IN intelligence and intelligence hosted either in applications at the edge of the PSTN or within the Internet itself.

In the case of interworking between IN and intelligence at the edge, the Internet provides a connectionless signalling path between, typically, an IN SCP and a point of intelligence within customer premises equipment (CPE). Consider, for example, a call centre application where calls may be routed to agents distributed across a number of geographically dispersed call centres. The introduction of signalling between the automatic call distribution (ACD) system deployed within the call centre and the IN SCP would enable IN call routing to be based on state information (e.g. number of agents available) received from

the CPE ACD as well as traditional IN routing criteria such as originating area and date/time-based routing.

In the second case, intelligence within the IN co-operates with intelligence embedded within the routing and naming fabric of the Internet itself. An example of this model of co-operation could be configuration information stored in an IN user profile that controls the use of Internet QoS mechanisms (e.g. RSVP) to provide a higher level of service.

#### 6.2 Influencing medium- and long-term IN evolution

A number of relatively short-term examples for exploiting interworking between the IN and the Internet have been considered above. Indeed, several telcos and equipment vendors are trialling services built on such interworking. It is also important, however, to take a step back and consider some of the broader implications of the Internet on the medium and longer term evolution of the IN. This section concludes with a brief consideration of some of these broader issues.

IN services will increasingly need to interact with non-IN services at the network edge. What are the implications for IN service creation in supporting creation of the non-IN aspects of this service and defining the interaction between IN and non-IN elements? Furthermore, this interaction typically requires proprietary extension to SCPs to provide service logic with access to IP capabilities. Such extension is counter to one of the key motivations underlying the IN concept, namely multi-vendor interworking.

More generally, it is useful to compare the orthogonal service models supported by IN and the Internet to identify where the integration or interworking of elements from each model may yield a more capable, dynamic and flexible service platform. Characteristics of the Internet that may provide 'steer' to future IN evolution include the following.

- Network evolution decoupled from service specifics through the deployment of generic capabilities and open interfaces — IN evolution builds very much on the definition of specific 'baseline services' and technology assumptions.
- Multi-protocol architecture with protocols optimised for messaging between specific functions and applications — IN operates a single application part protocol used for communication between all application service elements. Optimisation and rapid evolution within the single protocol model is more difficult.
- Loosely coupled standards architecture allowing effective prioritisation and rapid development of key capabilities — IN evolution is governed by a large

tightly coupled set of recommendations. As a consequence, it is difficult for individual elements within the framework to be developed or enhanced independently of the framework itself. While the need for standards is not disputed, there may be benefits in seeking a more flexible approach.

## 7. Conclusions

The Internet protocol suite represents a fundamental change in the role of networks, services and processes in the satisfaction of user requirements when compared to traditional telco approaches. The combination of the flexible computer CPE, the open and layered architecture of the Internet protocol suite and the deregulation of markets across the globe has led to an explosion in services and connectivity never experienced before in communications. This will be supported further as the costs of infrastructure falls both with the growth of the networks and the technological advancements in the CPE and network technologies. The convergence of telecommunications and computing at a time of major political programmes, such as the Global Information Infrastructure [15], now promises continued expansion and a bandwidth and service 'big bang'. It is ironic that the 'big bang' has occurred on the bomb-proof network!

Another stimulant in the market is the role of the content provider and advertiser as a subsidiser of the internetwork. This is inevitable as the Internet is a highly customisable and global channel for advertising, marketing and sales. Companies will connect with the Internet as an integral part of doing business. The connection/subscription revenues grow the network which brings economies of scale. The desire to be innovative in the presentation and functionality of company WWW pages will fuel bandwidth growth to support real-time information and communications services. Different users in the Internet will require different types of connectivity (bandwidth, leased/switched, technology, mobility, performance) depending on their ability to pay and the package of capability requirements. However, they will still want the same interface and basic capabilities such as e-mail and WWW over their particular access network technology. The interactions between Internet services and access/core networks will allow Internet providers to migrate users up the network capability chain in a similar situation to that utilised by PC hardware and software vendors. More powerful software pushes up hardware requirements and hardware upgrade opens up new service capabilities.

While profits will be limited in the short term, the opportunity for future profits as the Internet stabilises and the benefits of multiservice, internetwork connectivity accrue will be inevitable and addressable by those companies who can bring the connectivity and software together for mass-market end-to-end services. The revenues

may be associated with the sale of access to value-added network capabilities (multicast, mobility, intelligence, end-to-end QoS), generic service functional elements (browsers, servers, content facilities, gateways) and stable mass-market services offering performance guarantees to the user. In particular, the Intranet which embraces and requires this synergy between the connectivity and software for company IT solutions will be a key new market.

The Internet protocol suite can satisfy these diverse requirements because it is designed to adapt to, and utilise, the underlying network and supports high-levels of protocol reuse. Both legacy and future network technologies can be brought together under the umbrella of the Internet suite of protocols to support ubiquitous control, intelligence, QoS and service-delivery mechanisms which will continue to allow diverse user requirements to be met. The Internet represents a revolution in networking, is a promoter of connectivity and new services and is already a proven, if dynamic, market. The Internet protocol suite will therefore become the broadband multiservice network of the future by means of organic growth, the development, promotion and use of new and diverse network technologies and the natural demands of attached users. While the individual protocols will certainly evolve and be superseded, the realities of deregulation, *de facto* standards and information networking suggests that network futures will be based on internetworking and the IP architecture.

In fact, significant protocol design decisions are now evident for the industry, particularly for the next generation of IP (IPv6). The Internet has been growing at an exponential rate and addresses are running out. This was exacerbated by existing Internet address allocation policy in which applicants were allocated an address range which was not closely coupled to their requirements or the structure of the network. Routers are therefore also experiencing performance problems due to the number of routes which need to be advertised on the backbone and the need to perform hop-by-hop processing of certain packet fields. Commercialisation and mission-critical applications finally require the development of an efficient method to support encryption, authentication and non-repudiation at the network layer and to identify packets with negotiated QoS parameters.

While this paper attempts to identify some of the major architectural issues for the future of networking, it is neither exhaustive nor comprehensive in its coverage. For example, switched public networks provide global switched connections, leased-lines provide national and international point-to-point bandwidth and satellites can provide switched and unidirectional bandwidth which affect routing but provide significant broadcast and multicast capabilities. Which combination of technologies, where and when, provide the best environments for IP customer networking and what are the implications on the existing

Internet protocol suite of new access technologies such as ATM, cellular radio, ADSL and cable modems? And how will the synergy develop between network, internetwork and CPE-based intelligence and customer data? With so much uncertainty surrounding standards and technology, it is essential to be able to look towards companies for network solutions, with a depth of understanding of these major technical and evolution issues. Many open questions and pitfalls exist in addressing the internetworking revolution, but it is hoped that this paper has at least identified a number of very positive consequences of this revolution.

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He then spent two years on customer private ATM networks before helping CONCERT to design and implement the network testing of the joint BT-MCI voice network platform. He gained his BT Martlesham MSc from University College London in 1994 and since then has been developing a team of researchers in the area of Internet transport research. This is closely aligned to the work of the Internet Engineering Task Force and he is presently acting as the BT standards coordinator for this body. The team has particularly focused on the areas of TCP, multicast, reservation, integrated services and mobility. Convergence of IP/ATM and IP/IN continue to be significant study areas for the team as well as novel evolutionary scenarios for the elements of the Internet protocol suite.

